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1-C:

A – Point “A” represents the profit maximizing equilibrium because that is when the Marginal cost is equal to Marginal revenue.

B – Point “B” represents the equilibrium where ATC is at its minimum because Marginal cost is equal to Average total cost.

C – This is correct because point C represents the equilibrium where Marginal cost is equal to Average revenue (price), which indicates **allocative efficiency**

D – Point D represents the point where normal profits are made because Average revenue is equal to Average cost.

2 – C:

A and B – When government intervention results in positive externalities(option A) due to increased availability of merit goods (option B), this represents the success of government intervention.

C – When government intervention results in a less efficient outcome, this represents an argument against government intervention, as a result of which, option C, becomes correct.

D – When government intervention reduces information failure, this again represents the success of government policy.

3-C:

A government is choosing between four routes for a new road. The table provides details of the costs and benefits associated with each route.

The government wants to get the highest return for taxpayers in terms of the social benefit relative to the private cost.

Which route will the government choose?

	route (\$ million)			SB	SB/PC
	private costs	private benefits	external benefits		
A	20	25	5	30	1.5
B	25	30	10	40	1.6
C	20	25	10	35	1.75
D	30	35	10	45	1.5

value of SB/PC is the highest

→ The highest return in terms of the social benefit relative to the private case.

This is why, key C, is correct.

4 – D:

* Consumer equilibrium $\Rightarrow$

 satisfaction maximized

$$* \frac{MUP \cdot B}{P \cdot P \cdot B} = \frac{MUA \cdot B}{P \cdot A \cdot B}$$
$$* \frac{12}{3} = \frac{x}{6}$$
$$* \frac{12}{3} = 24 \text{ utils}$$

5- C:

A – It is in case of a normal good that an increase in price brings about a **contraction in qty demanded according to the substitution effect**. And, the income effect goes in the same direction, because when an increase in price causes a decrease in real income, it will lead to a **contraction in qty demanded of the normal good**.

B – This represents the result of a decrease in price of a normal good, as in such a case, both income effect and the substitution effect will result in a consumer buying more of the good, **this is because, the income effect and the substitution effect move in the same direction**.

C- In case of a rise in the price of the inferior good, there will be a **contraction in qty demanded according to the substitution effect**, and since the income effect and the substitution effect move in the opposite direction, a rise in price when brings about a fall in real income, results in an **extension in qty demanded of the inferior good**, according to the **income effect**. This is why, key C, becomes correct.

D - In case of a fall in the price of the inferior good, there will be an **extension in the qty demanded according to the substitution effect**, and since the income effect and the substitution effect move in the opposite direction, a fall in price when brings about a rise in real income, results in a **contraction in qty demanded of the inferior good**, according to the **income effect**.

6 – C:

A: Profit maximization occurs when “**marginal cost is equal to marginal revenue**”

B: Sales or units sold are maximized when “**normal profits are made**”, that is when a **firm charges the lowest price, it possibly can**.

C: Sales revenue is maximized when “**Marginal Revenue is zero**”. This is why, key C, becomes correct.

D: **Profit satisficing is an alternative business objective to maximizing profits. It means that the business is making enough profits to keep shareholders happy or it is sufficient for investors to maintain confidence in the management they appoint.** It does not occur when “marginal cost is equal to zero”.

7 – B:

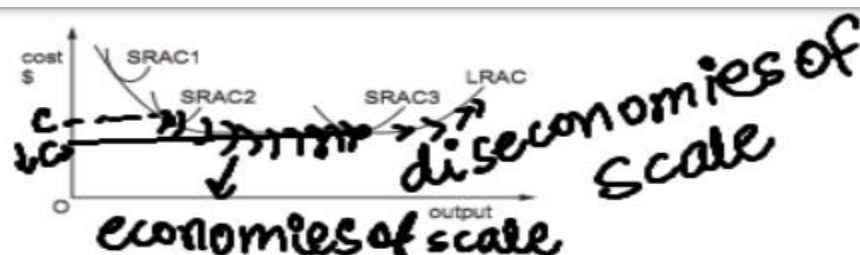
A – A cartel occurs when an oligopoly turns into a monopoly

B _ If an investment bank takes over a satellite communication company, this is an example of “**diversification**”. In this scenario, the investment bank, which traditionally deals with financial services, is expanding its operations into the satellite industry by acquiring a satellite company. By doing so, the investment bank is diversifying its business portfolio, as it now owns assets and operates in a different sector.

C- Lateral integration is a term that describes the action of merging two firms that deal in the sale of similar goods within one market branch. This combination is often caused by the company’s willingness to develop and expand its operations.

D – Vertical integration can be of two types. Vertical-forward integration occurs when a firm in the secondary sector merges with the one in the tertiary sector. And vertical-backward integration occurs when the firm in the secondary sector merges with the firm in the primary sector.

8- B:



A – When the minimum point of SRAC3 is lower than the minimum point of SRAC2, this means that a movement from SRAC2 to SRAC3 results in economies of scale, since it brings about a reduction in Long-Run Average costs.

B – This is correct as the law of diminishing returns applies in the short-run, and this explains why the SRAC curve is “U-shaped”. As more units of a variable input are added in the short run, initially, the total product (output) increases at an increasing rate. This is reflected by a steeply rising portion of the total product curve. **However, there comes a point where adding more of the variable input leads to smaller increases in output. At this stage, the marginal product starts to decline, indicating the onset of diminishing returns. The short-run average cost curve, which is derived from the total cost and total output, typically exhibits a U-shape. The upward sloping portion of the average cost curve reflects increasing average costs due to diminishing returns.** This occurs because additional units of the variable input lead to smaller increases in output, causing the **average cost per unit to rise.**

C- The position of the SRAC2 does not show the minimum efficient to scale position. This position occurs when the long-run average cost is minimized, and beyond which no further economies of scale can be exploited.

D – No, dis-economies of scale is shown through the rising part of the long-run average cost curve, that occurs after SRAC3. This can be seen in the diagram above.

9 – B:

Barriers to entry will lead to formation of a monopoly.

X-inefficiency happens when a lack of effective competition in an industry means that the average costs are higher than they would be if the market was more contestable. This leads to a loss of technical or productive efficiency.

A – When the firm does not account for external benefits, this represents under-allocation, an example of which, is the case of “merit goods”

B- High barriers of entry when lead to lack of competitive pressures , will result in X-inefficiency, that is when, the firm has no incentive to work hard, or cut its costs of production. This is why, key B, becomes correct.

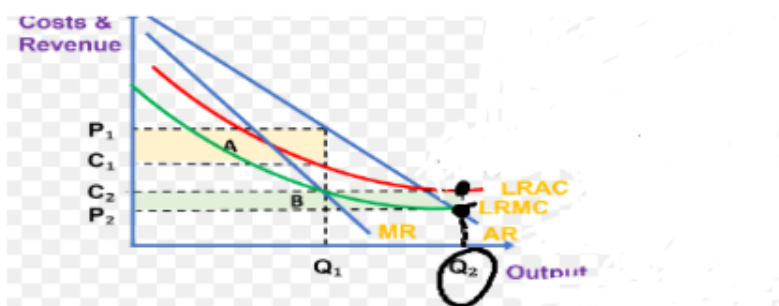
C and D:

Barriers to entry will lead to formation of a monopoly. This means that now , a monopolist will produce a profit maximizing level of output (where $MR=MC$).

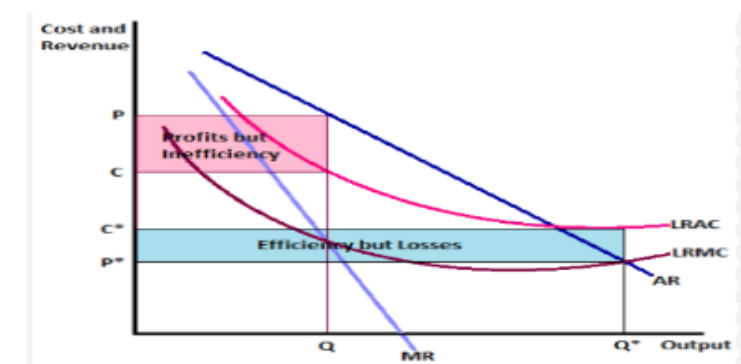
And this means that, the firm’s price set will be higher than its marginal costs or average costs. But, this is not the reason , which explains the existence of “X-inefficiency”

10- B:

A - In a natural monopoly, the industry exhibits **economies of scale**, where there are high fixed costs(as in case of utilities such as electricity whereby start-up costs are very high) but **falling average costs** as output increases. This means that as the firm produces and sells more units of the good or service, the average cost of production decreases. Due to economies of scale, a single firm can operate more efficiently and at a lower cost compared to multiple firms in the market. And, **full benefits of economies of scale** are exploited when a large volume of output is produced. This means that the minimum, efficient to scale point is achieved over a large output. This can be seen in the diagram below: (This in turn means that dis-economies of scale would not be experienced.)



B-



Since, long-run average cost is greater than the average revenue, this represents “per-unit loss”. And, when the per-unit loss is multiplied by the volume of output (Q^* - as can be seen in the diagram above), **which is allocatively efficient**(where $AR=MC$), this will help to determine the “**area of loss**”, that is seen in the diagram above. This is why, key B, will be correct.

C – This is incorrect because profits will only be maximized at the profit-maximizing level of output whereby Marginal cost(MC)= Marginal Revenue(MR).

D – This is incorrect too because sales revenue will be maximized when $MR=0$.

11- C:

- **The two leading brands control 60% of the market:** This indicates a **concentration of market power among a few dominant firms**. In an oligopoly, a **small number (4-5)** of large firms typically control a **significant portion of the market**. The fact that only two leading brands hold a substantial market share suggests an **oligopolistic market structure (AND NOT A MONOPOLY)**
- **Other large firms and a few smaller firms exist in the market:** **Oligopolistic markets** often have a **few large firms, known as oligopolists, dominating the market**. Additionally, there might be a **limited number of smaller firms trying to compete alongside the larger ones**.
- **Prices tend to be stable:** In an oligopoly, **firms are interdependent**, and their **pricing behavior often reflects stability**. The dominant firms in the oligopolistic market carefully observe and respond to each other's pricing actions, leading to **price stability within a certain range**. In case of a **non-collusive oligopoly**, **prices remain unchanged at the "kink" (shown through the kinked-demand curve model)** Price changes in oligopolistic markets are often strategic and coordinated to avoid price wars.
- There are significant barriers to entry: **Oligopolistic markets** tend to have **high barriers to entry**, which means **it is challenging for new firms to enter and compete effectively**. The presence of **significant barriers prevents new entrants from challenging the market dominance of the existing large firms**.
- This is why, key C, becomes incorrect.

A) Monopolistic Competition: In monopolistic competition, there are many firms (**20-25 small firms**) offering **differentiated products**, and each firm has **some degree of control over its price**. The concentration of **market power in the hands of two leading firms**, and **significant barriers to entry** represent that this market structure cannot be "**monopolistic competition**", because it is in case of a **monopolistic competition**, that they are less or no barriers to entry.

B) Monopoly: A **monopoly** occurs when there is **only one seller in the market, controlling majority of the supply of a particular product or service**. The presence of other large firms and the concentration of **market power in the hands of two leading firms**, as mentioned in the characteristics, rules out a **monopoly**.

D) Perfect Competition: **Perfect competition** is characterized by many small firms (**1000s of them, and each possesses a very insignificant market share**), selling identical products, **where no single firm has control over the market price**. The presence of dominant firms with significant market power, as indicated by the two leading brands controlling 60% of the market, contradicts the conditions for perfect competition. **Importantly, a perfect competition is characterized by "no barriers to entry"**

12 – D:

By definition , for “super-normal profits” to be earned, the Average revenue must be greater than the Average cost.

In both the cases ,where a profit-maximizing level of output is produced,(when **MC=MR**), and where sales revenue maximization has been done (when **MR=0**), supernormal profits are being made due to Average revenue being greater than Average cost, which can be **identified through the vertical distance between AR and AC.**

This explains why ,key D , will be correct.

13 – D:

A) Allowing private firms to merge and create a monopoly would be counterproductive in preventing overcharging. A monopoly would give a single firm complete control over the market, enabling it to set prices without competition or accountability. This would lead to higher prices **as the firm showcases its producer sovereignty.**

B) Appoint a regulatory body to guarantee supplies at all times will not lead to producers being prevented from overcharging.

C) Deregulation can lead to a free market, but it may not necessarily promote competition if a few large firms dominate the market. Without adequate regulation, firms may exploit their market power to overcharge consumers

D) Enabling customers to switch easily between providers is a crucial measure to prevent overcharging. By promoting competition, firms are incentivized to offer fair prices and quality services to retain customers or else **customers will switch to competitors because the demand is price-elastic.** Easy switching creates a market-driven incentive for firms to serve consumers efficiently. This promotes a competitive market where firms compete for customers, leading to better services and prices.

14-D:

A – Since firms within the scheme are required to have a sufficient number of permits to cover the amount of pollution generated in production, they will have to purchase the permits, and this potential rise in costs will encourage them to reduce environmental damage, so that they are able to reduce their costs of production. This represents a success of the policy

B -As a result of introduction of pollution permits, firms will be encouraged to improve their technology and adopt greener technology, so that they are able to reduce the number of permits they require, which will result in a reduction in their costs of production. This represents a success of the policy.

C – Firms will be incentivized to pollute less, so that they can sell some of their permits to other firms, which again represents a success of the policy.

D – If the permits do not impose strict enough limits on carbon emissions, this will result in the policy to fail.

15 – B:

Reduction in gini coefficient represents a reduction in inequality.

A regressive income tax system is one where lower-income individuals pay a higher proportion of their income in taxes, while higher-income individuals pay a lower proportion. **This is because not only does average rate of tax fall upon a rise in income, but also, the marginal rate of tax is lesser than the average tax, which represents that for every additional \$1 earned, less is taken in tax.** This would increase income inequality, leading to a higher Gini coefficient, which measures income inequality.

The other options are more likely to reduce the Gini coefficient:

A) Decreasing ad valorem sales tax rates which is an indirect tax will reduce the gini-coefficient because indirect taxes are regressive in nature.

C) A more progressive income tax system would increase the tax burden on higher-income individuals, reducing income inequality.

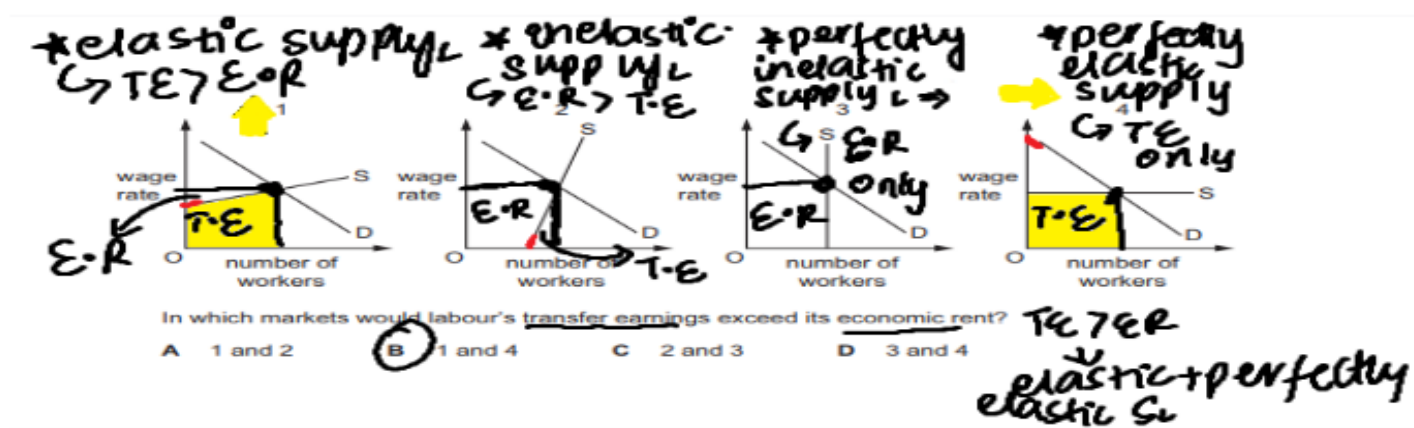
D) Increasing transfer payments to the unemployed would provide financial support to those in need, reducing poverty and income inequality.

16-C:

Marginal revenue productivity is $MPP \cdot MR$

C – The extra output produced by an additional worker represents the **marginal physical product of labor**, which when multiplied by **marginal revenue** gives **Marginal revenue productivity** as the answer. This explains why, according to formula, no other option can be correct.

17-B:



18- A:

A) This objective is not suitable for sustainable economic growth. Using all natural resources to achieve the highest growth rate would lead to overexploitation, environmental degradation, and unsustainable practices. This is why, this becomes correct.

B) This objective is suitable for sustainable economic growth. Careful management of non-renewable resources ensures their efficient use and conservation, minimizing waste and promoting sustainability

C) This objective is also suitable for sustainable economic growth. Protecting the environment, wildlife, and landscapes preserves natural resources for future generations and maintains ecosystem services essential for human well-being.

D) Meeting the needs of both current and future populations ensures that development is sustainable, considering the well-being of all generations.

19-C:

The correct answer is C) net property income from abroad.

Gross Domestic Product (GDP) and Gross National Income (GNI) differ in their scope:

- GDP measures production within a country's borders.
- GNI measures production by a country's citizens, regardless of location.

GNI equals GDP plus **net property income** (income receipts from abroad minus income payments to abroad)

For example, if a Japanese firm operates in the UK and sends profits back to Japan, this counts towards UK's GDP but not its GNI. Conversely, if a UK firm operates abroad and repatriates profits, it counts towards UK's GNI but not its GDP .

20-A:

A) In this case, the government official is using their position to take a bribe from someone seeking special treatment, rather than providing a legitimate service or creating value. This is an example of redistribution of existing wealth rather than creating new wealth. **This is why, this becomes correct.**

B) This is an example of earning income by renting a field(land) rather than seeking a share of existing wealth without creating new value.

Option C) Landlord being paid for renting a room is a legitimate income from providing a service (accommodation), and not an example of seeking a share of existing wealth without creating new value. It's a fair transaction between landlord and renter.

D) While the firm may be criticized for paying low wages, the entrepreneur is still creating new wealth by setting up a business and providing employment opportunities.

21-B:

According to the international labor force organization, unemployed are those who are willing and able to work at the current wage rate but are not able to find paid work. They are **actively seeking work** and are available to work within the next two weeks.

- **Working age population(16-65) includes:-**
In labor force (employed+ unemployed) + not in labor force.

This is why, everyone of working age who is not employed is not considered as unemployed because the working age also includes economically inactive group, who is not in labor force.

Moreover, since pension age people are “willing and able to work”, they will be classed as unemployed.

Additionally, since part time employees are “actively seeking full time employment”, they will also be classed as unemployed.

As a result of which, key B, is correct.

22-B:

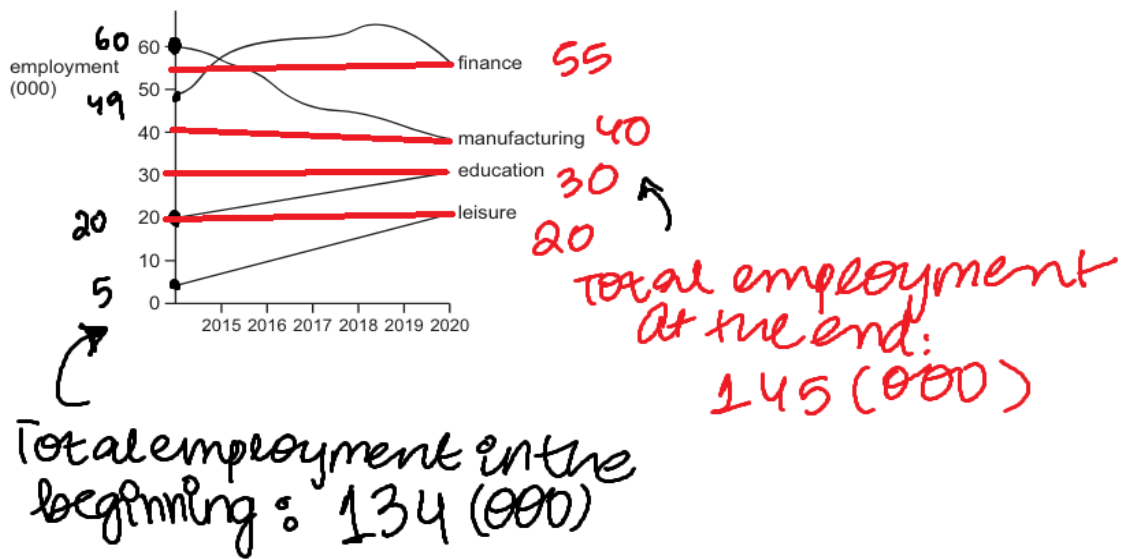
A – This is incorrect because up until 2015, **manufacturing** was the **biggest employer**.

B – Leisure had the fastest growth rate in employment because its slope is the steepest, which shows that the **employment rate increased at an increasing rate. That being said, it becomes correct.**

C – No, manufacturing was not the only industry to show signs of decline ; instead, finance also showed signs of decline.

D – Total employment in all four industries increased over the period.

The diagram shows changes in the numbers employed in four different industries in a country between 2015 and 2020.



23- A: Actual economic growth involves movement of a point inside of the PPC (i.e. the actual) towards the potential (i.e. the point on the boundary of the PPC)

In case of option A, the actual (shown through point S) moves closer to the potential (i.e. shown through point T)

This explains why key A will be correct.

B – Here, there is an inward shift of the PPC, representing that the productive capacity has declined.

C – Here, there is a rightward shift of the PPC from PPF 1 to PPF2, which represents potential growth.

D – This shows a recession, as the actual is moving towards the origin.

24-B:

Change in AE =

$$\rightarrow +4000(G) - 1500(I)$$

$$\rightarrow +2500$$

Value of the multiplier:-

$$1 / (mps + mpt + mpm)$$

$$1/(0.1+0.2+0.1)$$

→ 2.5

Change in AE * multiplier = Change in RNI

2500*2.5 -> 6250.

This is why, key B, is correct.

25- A:

A) When an outflow of currency (money leaving the country) becomes an inflow (money entering the country), it increases the money supply in the economy.

B) Higher cash-to-deposits ratios mean commercial banks can lend less money as they have to hold a greater proportion of deposits in cash form, reducing the money supply as now less new money will be created through the process of credit creation.

C) A government budget surplus means the government is taking in more revenue than its spending, reducing the amount of money in circulation.

D) Borrowing from the public (e.g., through bonds) will also transfer money from the public's hand to that of the government.

26- D: An increase in the precautionary motive for holding money should increase money demand, i.e. demand for liquidity.

Because of this the liquidity preference curve will shift rightward from LP To LP1. This makes key D correct.

A – Here, the money supply is shown to increase.

B – Here, the money supply is shown to decrease.

C – Here the liquidity preference is shown to decrease.

27-B:

The scenario describes a bilateral aid arrangement between the UK and Uganda, where the UK provides aid directly to the Ugandan government.

The condition that the materials must be bought from the UK makes the aid "tied aid", which means the recipient country is required to spend the aid on goods or services from the donor country.

Since the aid is for a specific project (building a school), it is also an example of project aid.

Multilateral aid, on the other hand, involves multiple countries or international organizations working together to provide aid, such as through the United Nations or the World Bank. In this scenario, there is no multilateral aid taking place.

So, the correct answer is B) multilateral.

28-D:

D) When unemployment increases, the government's revenue from taxes decreases, and its spending on unemployment benefits and social support programs increases. This leads to a larger budget deficit.

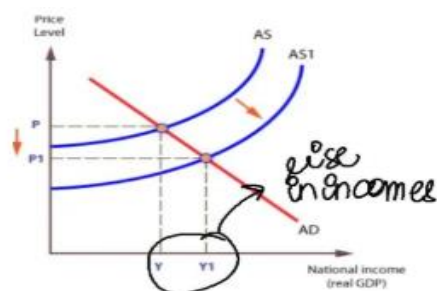
A) Unemployment doesn't directly lead to a switch from direct taxation to indirect taxation.

B) Unemployment is typically associated with decreased Aggregate demand, this is because a fall in incomes result in a fall in consumption expenditure. This in turn leads to a reduction in demand pull inflation.

C) Because of unemployment when the incomes of individuals fall, there is a decrease in import spending which leads to a fall in import expenditures, thereby resulting in a decrease in Balance of trade deficit.

29- D:

If the government of Lesotho provides subsidies to local entrepreneurs, this will have the effect of increasing the government expenditure. Moreover, since provision of subsidies is a supply-side policy, it will reduce the costs, and increase Aggregate Supply, which will have the effect of an increase in incomes from Y to Y_1 , as can be seen in the diagram below. However, although this makes exported goods competitive, whether or not the Balance of trade position improves, depends upon the price elasticity of demand. And also, there are other components of current account as well. This means that the current account position will be uncertain.



* X prices $\downarrow$	* X prices $\downarrow$
X Qty $\uparrow$ $\rightarrow$ elastic demand	X Qty $\uparrow$ inelastic demand
$X(CTR) \uparrow$	$X(CTR) \downarrow$
BOT position improves	BOT position worsens

30 – B:

